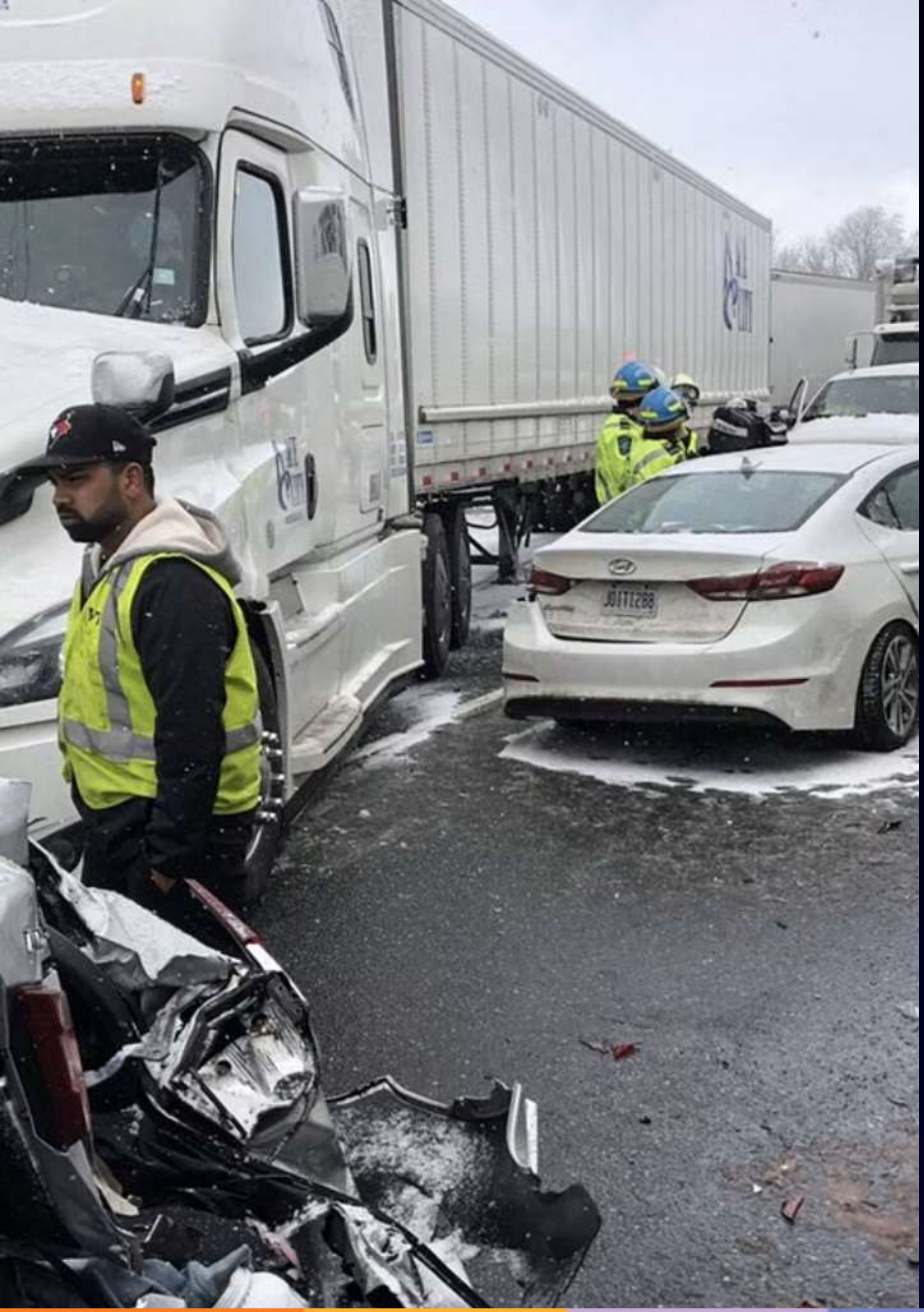

Safe Roads Competition 2024

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1. Executive Summary

After analyzing different datasets involving collision, fatalities, vehicle population among others as well as technical studies and articles, we proposed several solutions to mitigate the number of collisions in the city of Toronto.

Advanced data analysis techniques and state-of-the-art statistics software provided by SAS are used in the study.

2. Objectives

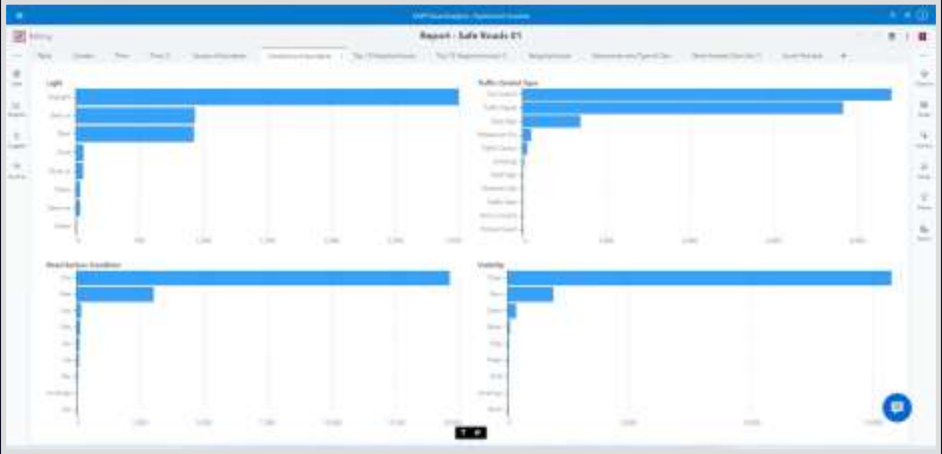
Propose solutions to decrease the collisions and fatalities in the city of Toronto, improving road safety for all.

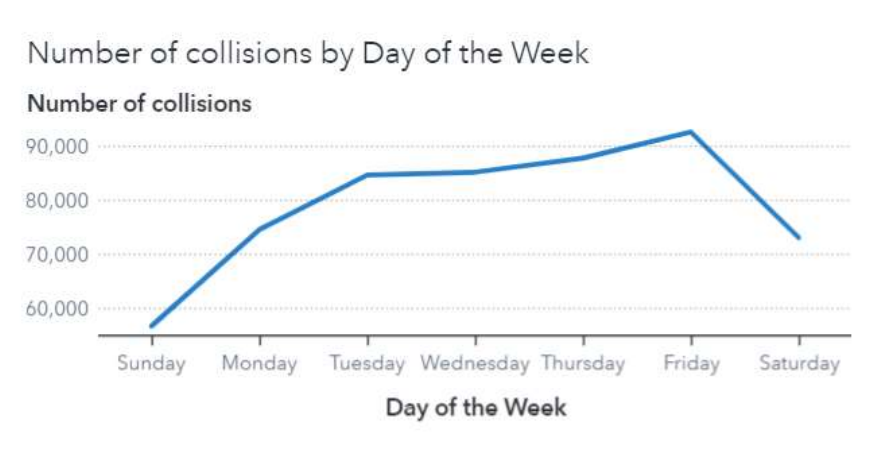
3. Background

- Each year in Canada:
    - Over 1,800 people are killed
    - Over 150,000 are injured (including 10,000 seriously)
    - Costs society $40.7 billion annually (2.1% of Canadian GDP)
- Improving road systems can reduce fatalities and injuries
- Distracted driving defined as diverting attention from driving task
- About 2% of fatal and injury collisions involve mobile device use

4. Data Analysis

Overview of Datasets Used for Analysis





Comprehensive Data Collection

Various datasets containing detailed information on collision incidents.

Insightful Reports

Reports with valuable insights into the nature and frequency of collisions.

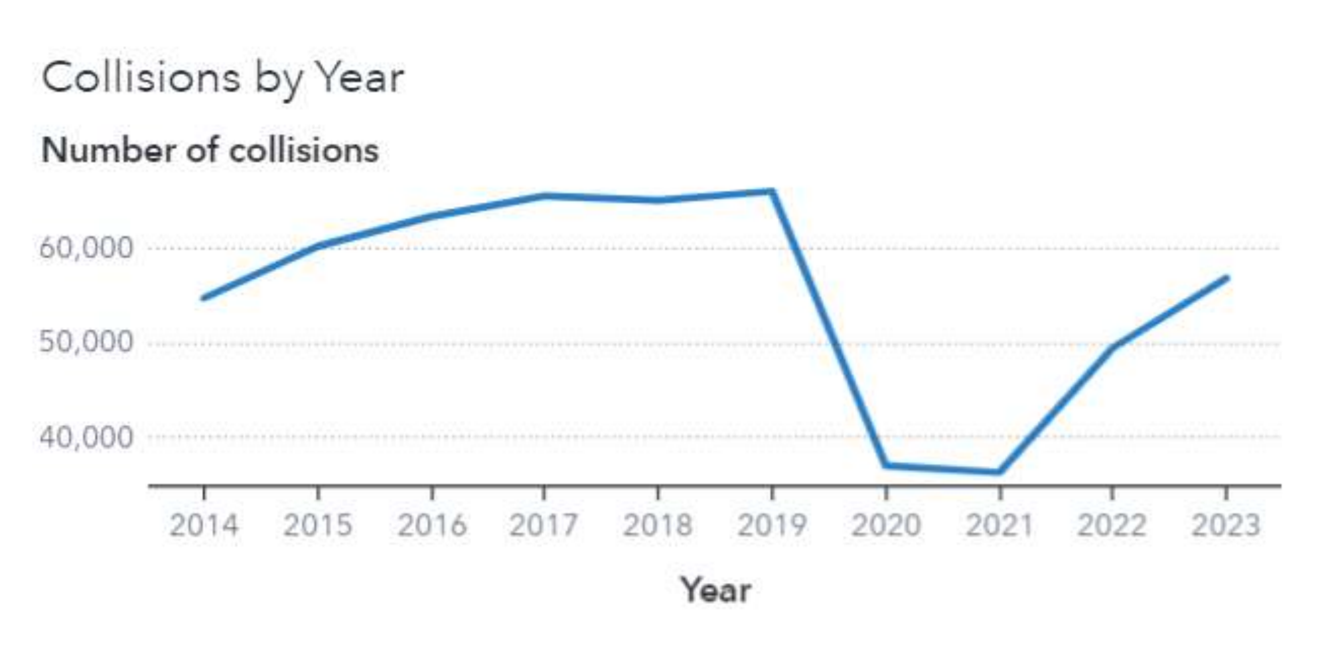
Visual Data Representation

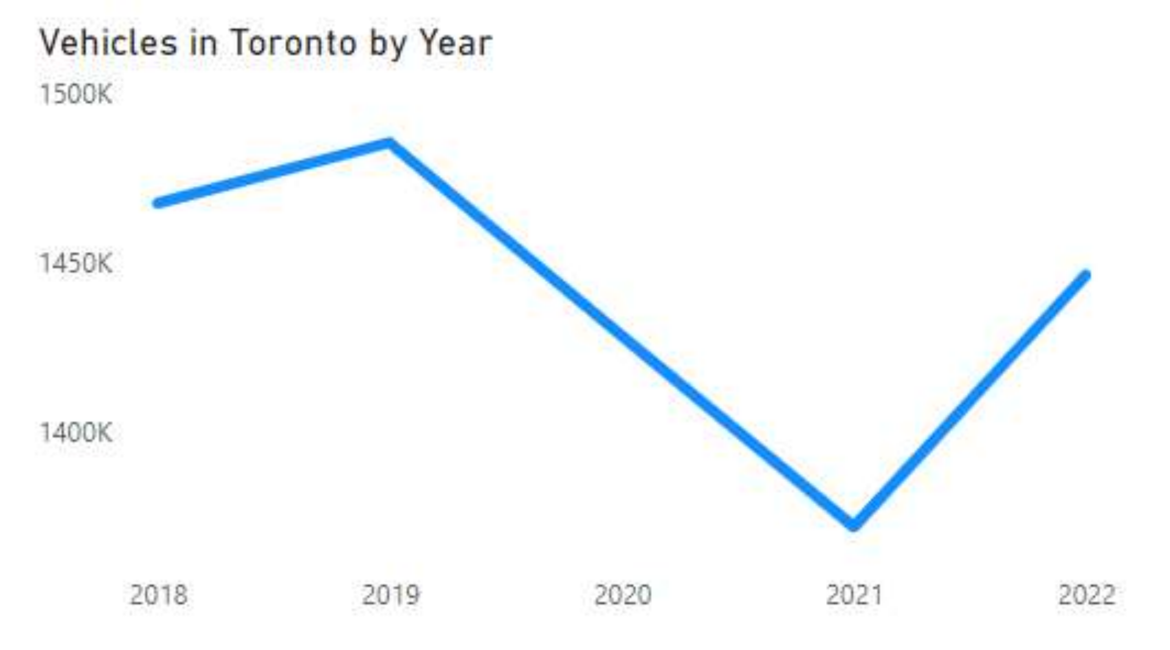
Charts and graphs illustrating the collision trends for better understanding.

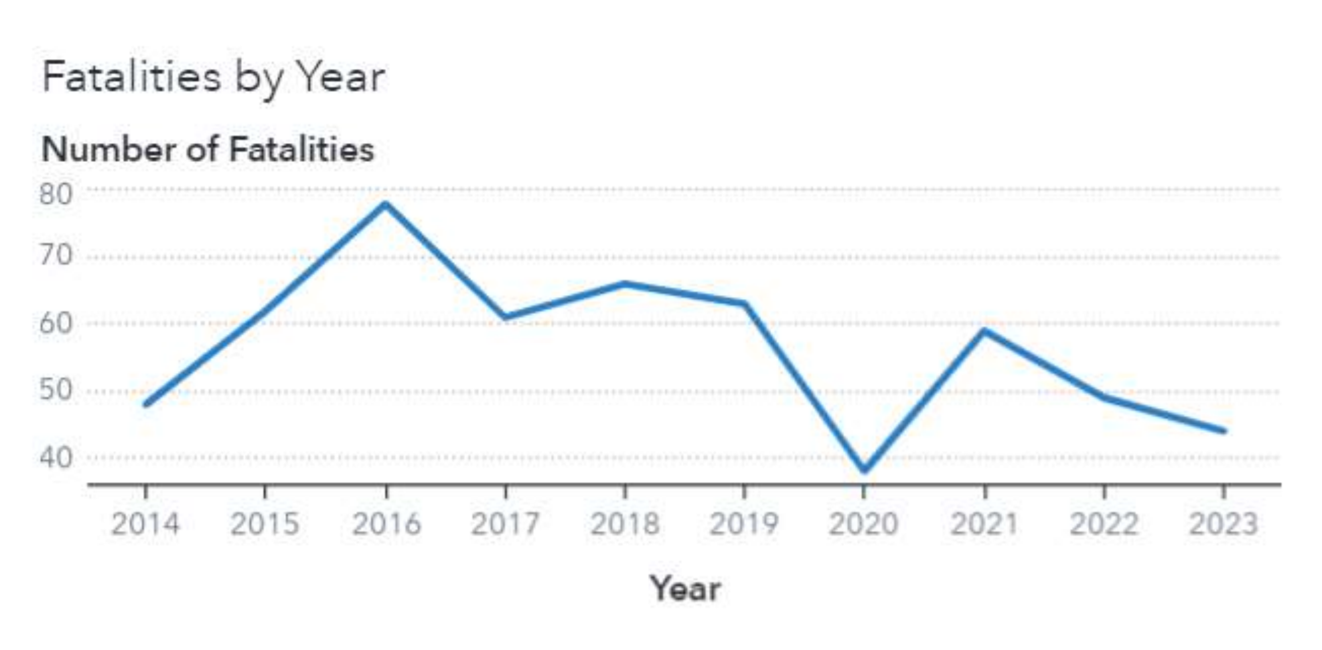
4.Data Analysis

Yearly Analysis on Collisions and Fatalities

Collisions follow approximately the same trend as the vehicle count in Toronto, however the fatalities have been decreasing (except for 2020 pandemic effect).







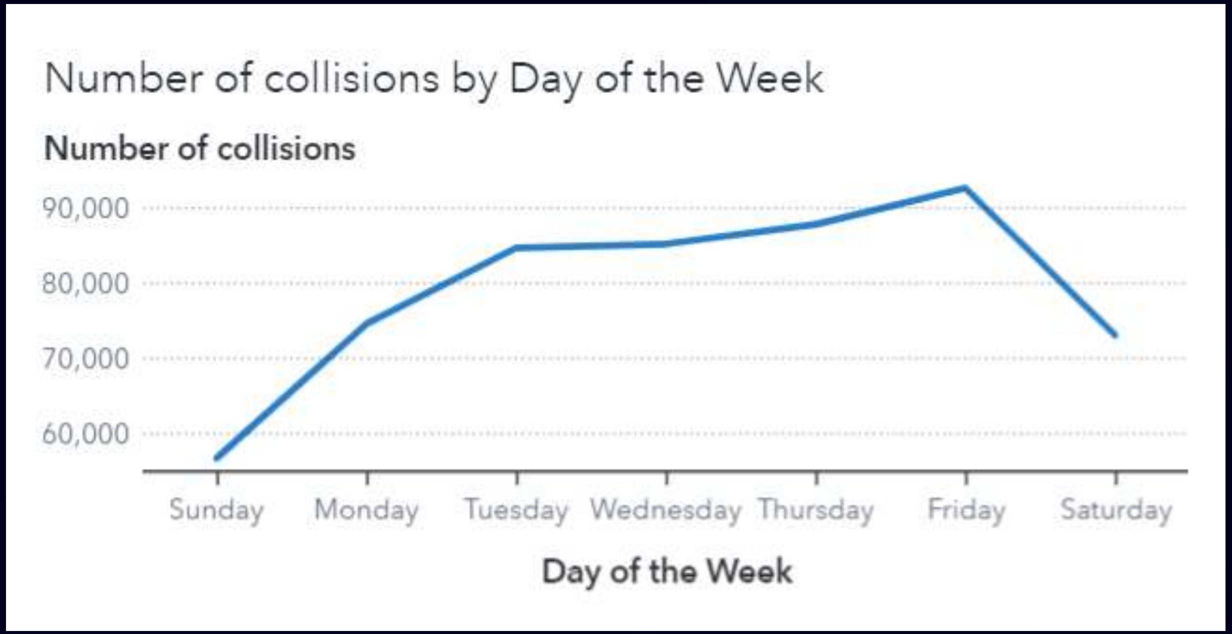
4. Data Analysis

Occurrences by Dates and Times

Most collisions occur on Thursdays and Fridays between 3pm and 6pm.

November, October and December are the months more prone to accidents.

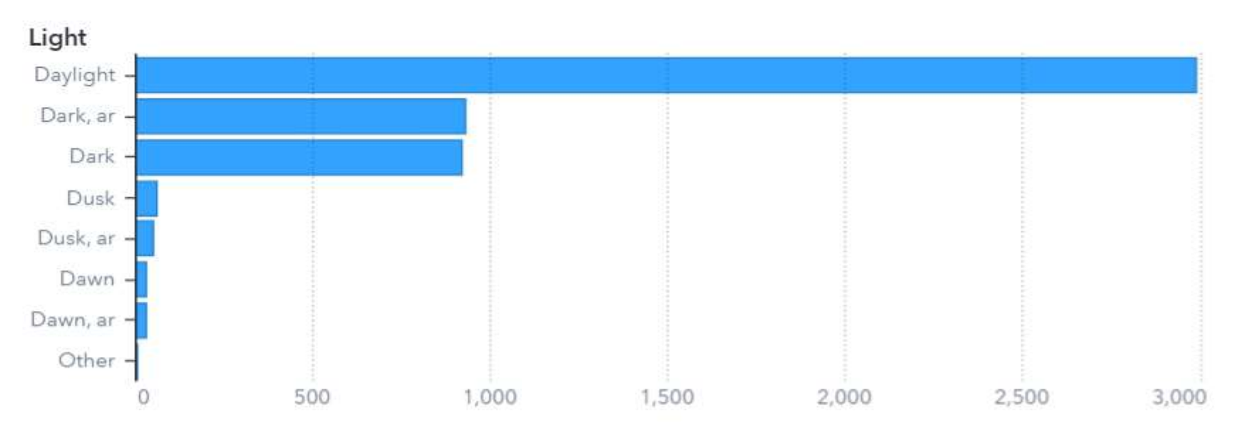
Month	Collisions	Fatalities
Novembe	50,386	59
October	49,760	51
Decembe	49,020	52
Septemb	48,080	61
June	47,530	40
January	46,824	47
August	45,936	60
July	45,864	55
Februar	45,565	27
May	45,350	46
March	42,456	36
April	38,537	34

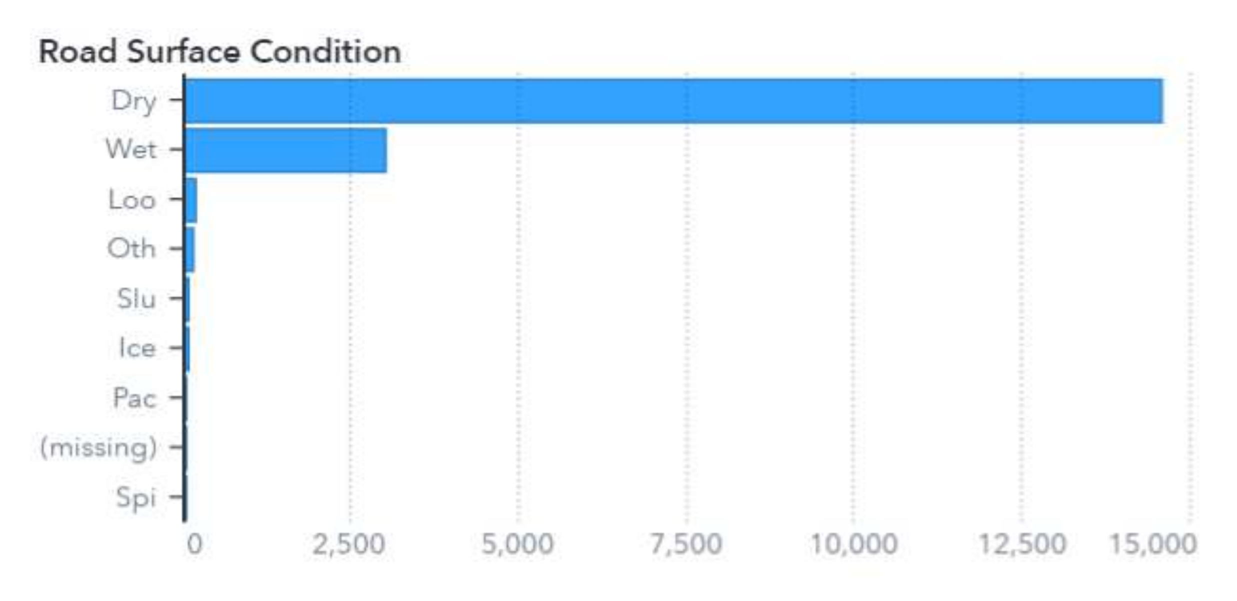


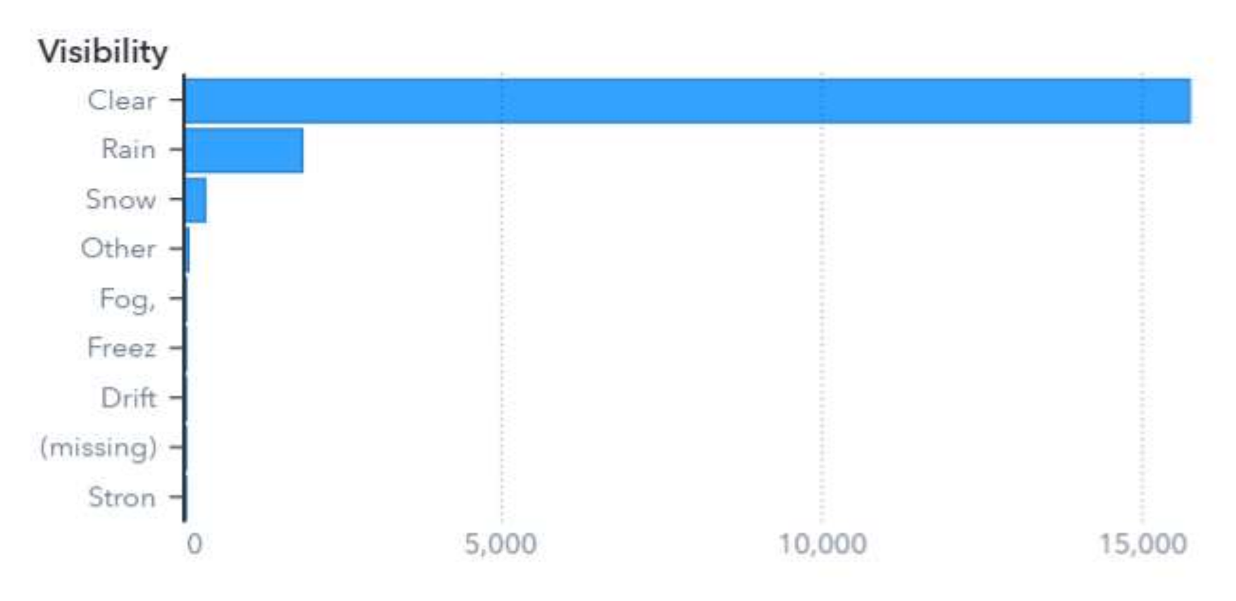
4. Data Analysis

Accidents Conditions

The typical collisions occurs with daylight, clear visibility and dry road conditions



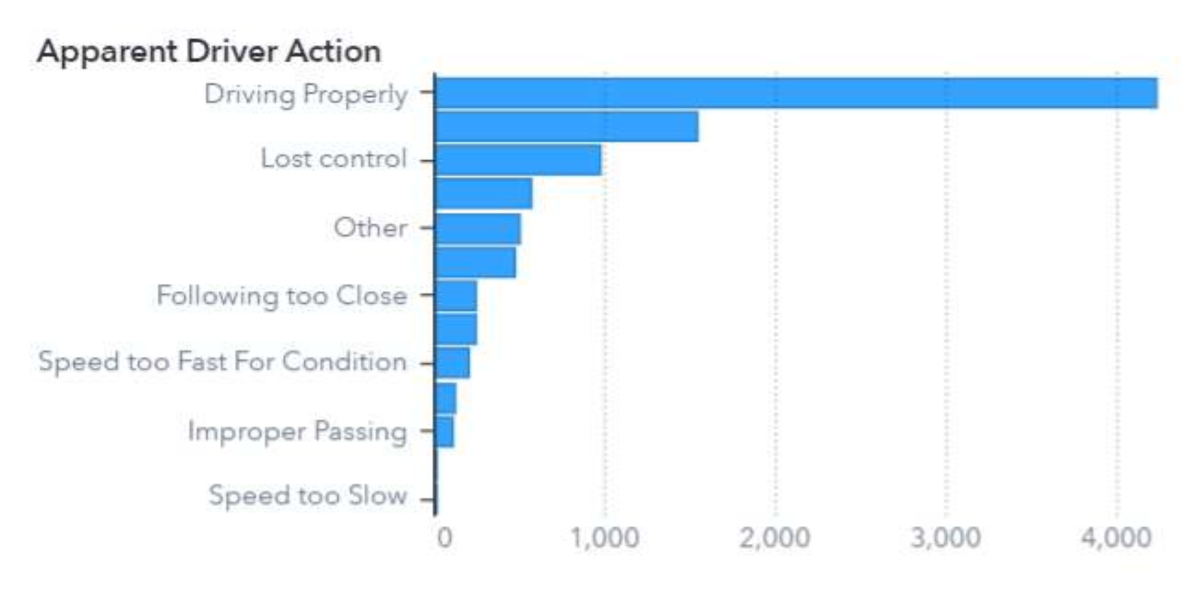


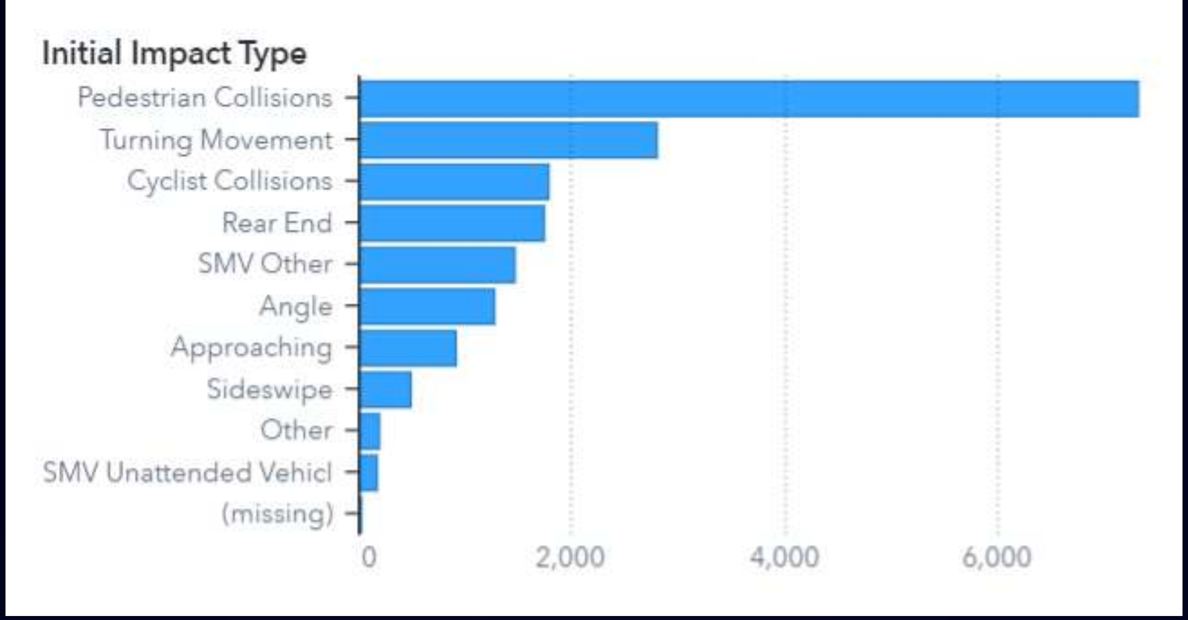


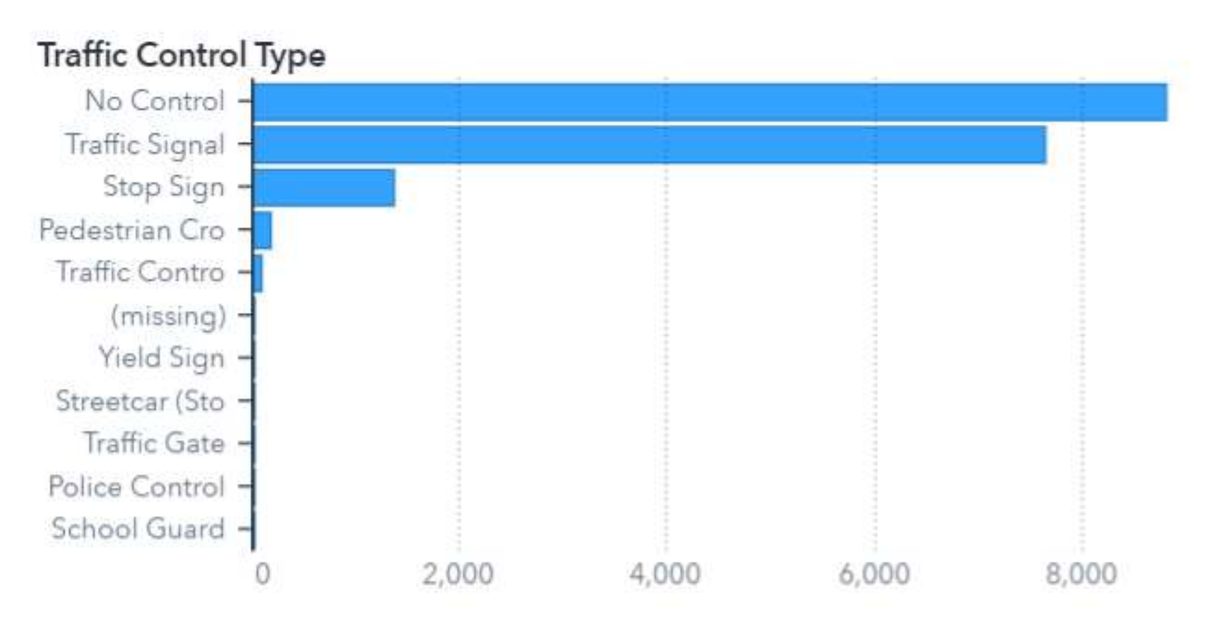
4. Data Analysis

Understanding Traffic Accidents

In more than half of the collision the driver was apparently driving properly.



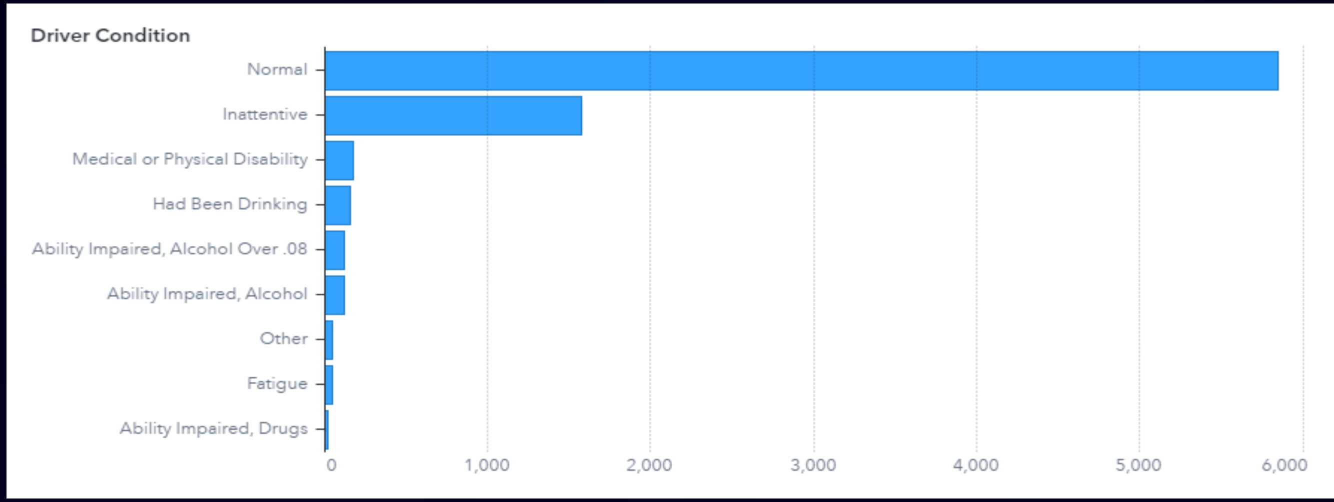


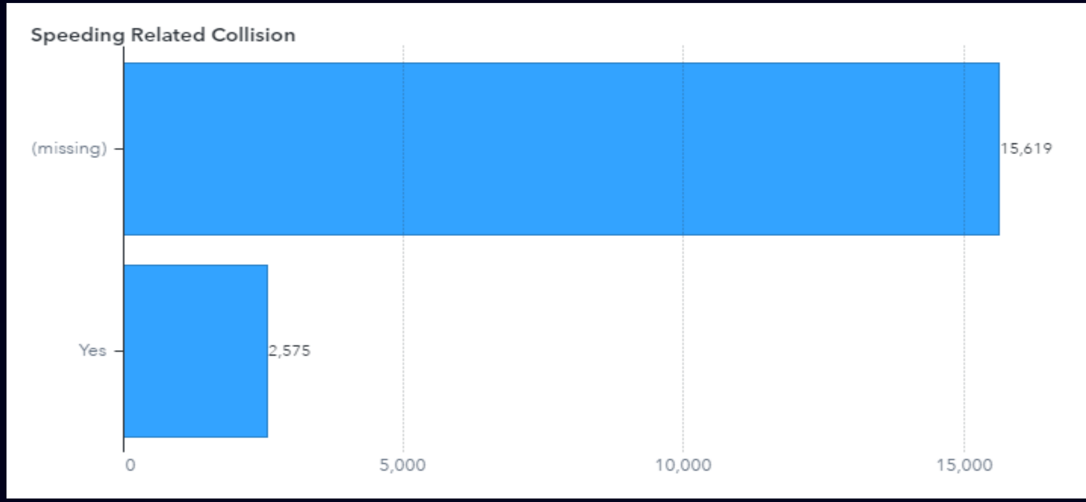


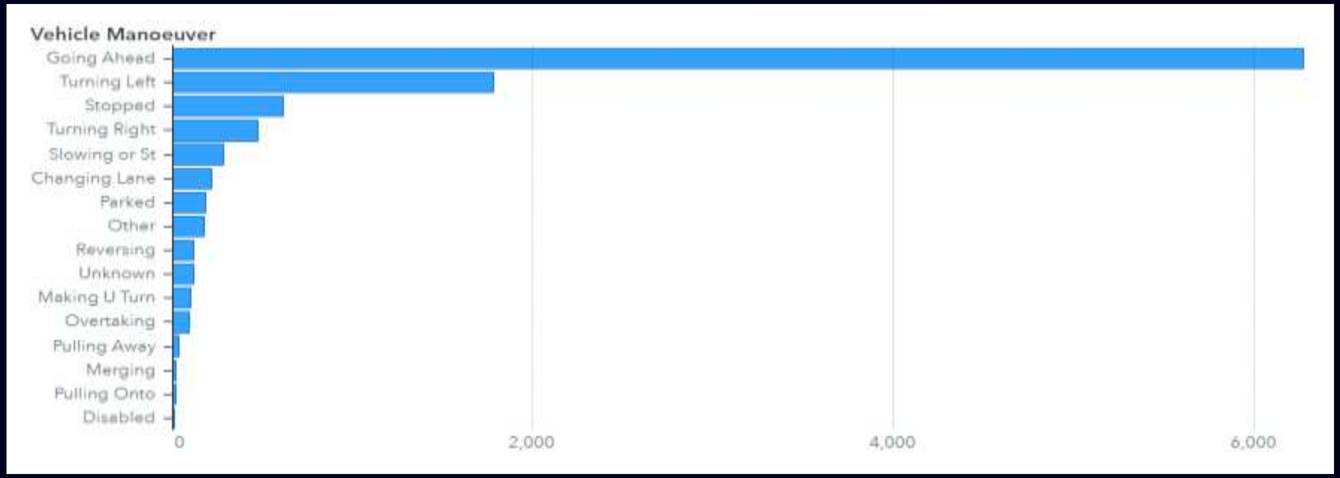
4. Data Analysis

Understanding Traffic Accidents

The driver was typically not impaired, going ahead and reported speeding more than than 20% of the time.

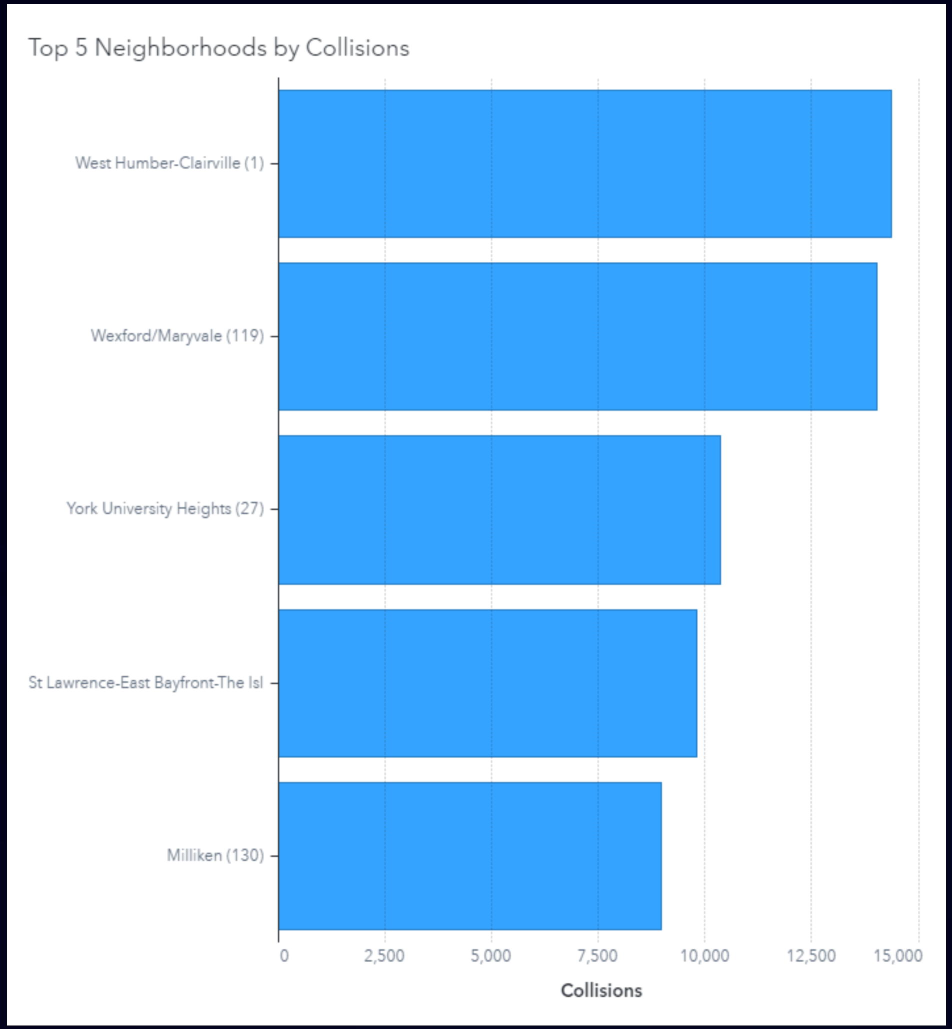






4. Data Analysis

Top 5 Neighbourhoods by Collisions



- We will focus on the Top neighbourhoods in which most collision occur.
- They follow the same temporal and descriptive pattern as the typical accident.

Neighbourhoods:

1. West Humber-Clairville
2. Wexford/Maryvale
3. York University Heights
4. St Lawrence-East Bayfront-The Islands
5. Milliken

5. Insights

Key Insights and Trends Identified from the Analysis

1

High Incidence Areas

Identification of Top 5 neighbourhoods with the highest frequency of collisions.

2

Peak Collision Times

Revealing the most common times for collisions to occur. Thursday and Fridays from 3:00PM to 6:00PM.

3

Visibility and Road conditions

Daylight, clear vision, dry road, going ahead, and apparently driving properly

4

Primary Contributing Factors

Analysis of key factors that lead to collisions. Distracted drivers, fail to yield, and speeding.

6. Solutions

1. **HIVE Implementation**
   Enhancing road infrastructure to minimize collision risks.

2. **Educational Campaigns**
   Initiating campaigns to raise awareness about safe driving practices.

3. **Zone guard Flexible Delineators**
   Focused on speed limit and yield

4. **Yield signs**
   Visually striking reinforcing current signs

5. **Fine Amount Sign Boards**
   Strengthening enforcements whilst serving as a deterrent

6. Solutions

HIVE Implementation

- High Visibility Enforcement: Redistribute the police patrol from low collision areas to high collision areas on Thursdays and Friday from 15:00 to 18:00.

- The temporary increased police presence and enforcement during peak collision times will deter risky driving behaviours.

- Proven to work in other cities like Arizona with 27% reduction in collisions.

6. Solutions

360º Safe Roads BTL Educational Campaigns

- Use social media, and highway teleprompters for awareness on collision factors.
- Implement traffic education workshops for senior high school students in top collision neighbourhoods and for all Colleges and Universities in the City of Toronto.
- Implement the Distracting Driving Awareness month.

Implementing visually striking yield signs along roads offers a swift and impactful strategy for minimizing collisions.

6. Solutions

YIELD signs painted on pavement reinforcing current signs

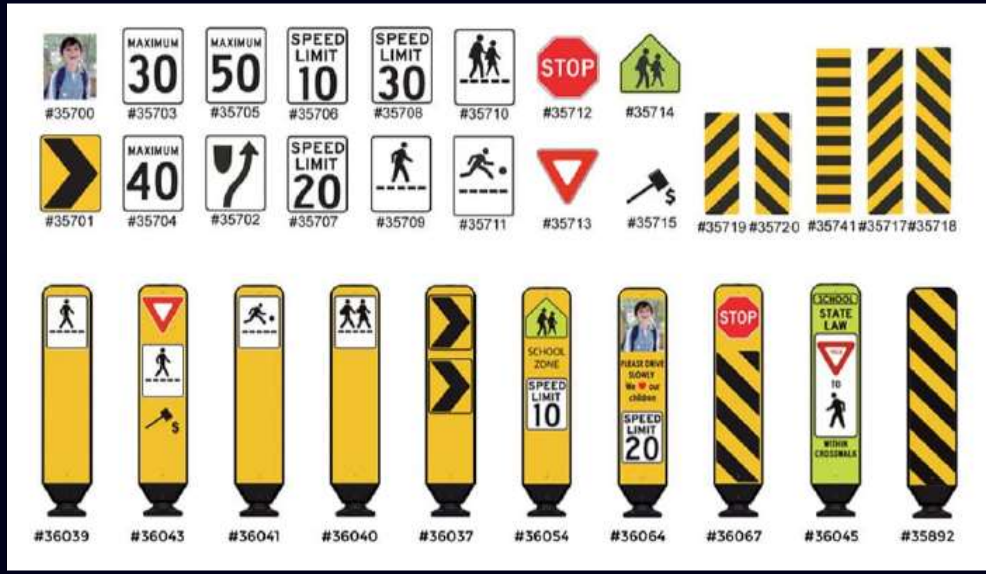
- Implementing visually striking yield signs along roads offers a swift and impactful strategy for minimizing collisions reinforcing current signalling.
- Add a radiant strip in shape of yield sign for clear vision at night.
- Relatively inexpensive to implement.

Cost Factor	Estimated Cost
Materials(Paints)	$4 - $5 per sign
Installation Labor	$100 per sign
Total Estimate (Initial Cost)	**$105 per sign**

6. Solutions

Zone guard Flexible Delineators Traffic Panels in critical zones

- Focused on speed limit and yield
- Provides the sensation of narrow lanes (making slow down).
- Information on guidance signs had a substantial influence on driving behaviour.
- Improve road safety in critical Zones.
- Easy/Fast Installation.



6. Solutions

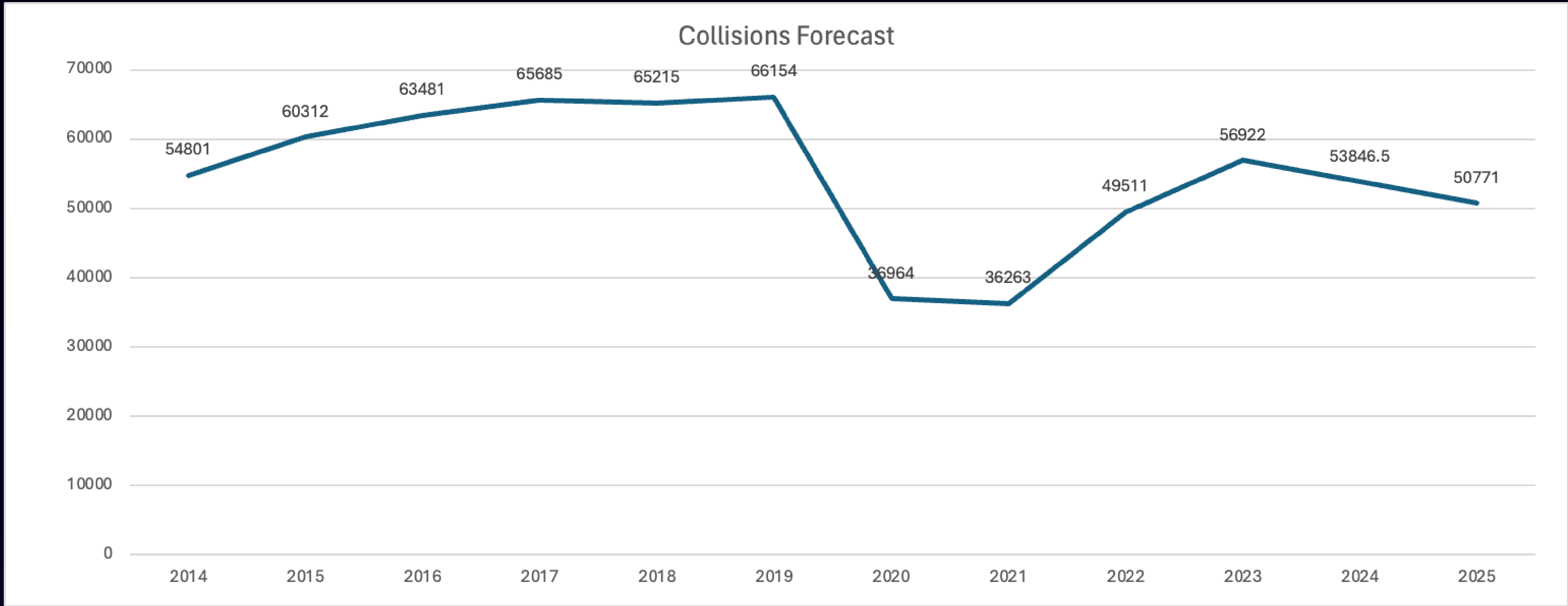
Install Fine Amount Sign Boards

- Fine amount reminder signs promote compliance and responsibility.
- They prevent collisions and encourage safer driving behaviours.
- These signs raise awareness and promote cautious driving (Behavioural influence).
- Safer practices lead to fewer collisions and safer roads.
- Fine amount reminders contribute to overall road safety and orderliness.

Cost Factor	Estimated Cost
Board (48’’x24’’)	$400 per sign
Installation Labor	$200 per sign
Poles	$150 per sign
Total Estimate (Initial Cost)	**$750 per sign**

Forecasting

We forecast a reduction of collisions in top 5 neighbourhoods by 12.1% after implementation of our solution, align to City of Toronto Vision Zero Plan.



Conclusion

Our analysis reveals that traffic collisions account for 2.1% of Canada's GDP. We've identified speeding, distraction, and failure to yield right of way as the primary causes of these collisions.

To address this, we propose implementing five targeted solutions in affected neighbourhoods, aiming for a 12% reduction in collisions, thus aligning with the City of Toronto's Vision Zero 2025 agenda.

Furthermore, we suggest conducting further analysis, including examining traffic ticketing data, adjusting police patrol shifts, and gathering more precise information on accident conditions currently listed as unknown or missing data, to enhance the effectiveness of our interventions.

THANK YOU

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